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# X-Band Resonator for Studying the Various Axially Symmetric Inhomogeneities in the Frequency Band

I. Ivanchenko, M. Khruslov, N. Popenko, V. Plakhtii, S. Mykhaliuk
O. Ya. Usikov Institute for Radiophysics and Electronics
of the National Academy of Sciences of Ukraine
Kharkiv, Ukraine
ireburan@yahoo.com

***Abstract* — The problem related to the choice of an adequate microwave resonator for measuring the dielectric constant of substances with the high permittivity and losses is considered. Numerical modeling and experimental investigations of a rectangular X-band resonator with the axially symmetric inhomogeneities, such as a dielectric cylinder and the dielectric capillary filled with the solutions, which have the different dielectric permittivity, are carried out. Both optimal sample sizes depending on their dielectric permittivity and the geometric parameters of dielectric capillaries for studying the various liquids are determined.**

***Keywords — resonator, field distribution, dielectrics, solutions, permittivity***

## I. Introduction

The knowledge of regularities in changing the dielectric constants of various substances in a specific frequency band can be very important. There are many techniques developed to this end. In those situations when the changes are minor, the small perturbation method is the most effective for the measurements of aforementioned characteristics in the microwave range. Here, we are talking about the resonance methods of measurements applied to the dielectrometry problems. In such methods the material under test is inserted into the resonator and the permittivity and loss factor are determined on shifting the resonance frequency and changing in the Q-factor [1-3]. For the study of cylindrical samples the small holes are drilled in the resonator walls to insert the sample in the resonance volume with the help of the special holder [4, 5]. The situation becomes more complicated when it is necessary to measure the dielectric constant of substances with the high permittivity and losses, in particular, various aqueous solutions [6]. In this respect, the particular interest is the microwave range where there is a frequency dispersion maximum in the complex dielectric permittivity of many aqueous solutions, for example, the solutions having a biological nature. In this case, the experimental error should be no more than 0.1% or higher for the reliable identification of aqueous solutions of chemical and biological agents, and the differences between them. Moreover, one should consider the fact, that here it is necessary to deal with the small volumes of liquid samples.

When measuring the samples with a high dielectric permittivity and large losses, one of the problems is to minimize the effect of the sample on the characteristics of the resonator while maintaining a high sensitivity and correctness of measurements. In this regard, it becomes very important the problem concerning the choice of the optimal sample sizes or the geometric parameters of the dielectric capillaries for the investigation of various liquids.

In this paper, the study of a rectangular X-band resonator with the axially symmetric inhomogeneities, such as a dielectric cylinder and the dielectric capillary filled with the solutions having the different dielectric permittivity is carried out with the aim to create the adequate resonator for measurements the dielectric permittivity.

## II. Description of Measuring Module

A rectangular X-band resonator with the length L =160mm was manufactured and used as the measuring module (Fig. 1). The end walls of the resonator are the copper diaphragms of thickness h = 0.3mm with the central holes of diameter d1 = 8mm.

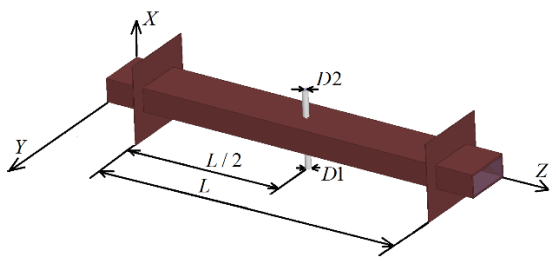


Fig. 1. Measuring module.

Since one of the ways to improve the quality of measurements of the dielectric constant is to obtain its value not as a result of a single measurement, but as a result of processing an array of data, in this case we assume the use of the resonance frequency dependence on the sample position. To implement this approach, it is necessary to include in this study an analysis of the influence of the samples on the spatial field distribution along the resonator axis. In this regard, in the central part of the resonator the longitudinal slot in both wide walls of the resonator was cut for the possibility of changing the position of a cylindrical sample under test. The slot length is 30 mm and its width is equal to the diameter of the sample or the external diameter of the dielectric capillary.

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## III. Numerical Simulations and Experimental Study

Numerical simulations of the characteristics of a resonator with the cylindrical samples of different permittivity and a resonator with the Teflon capillary filled with the solutions of different permittivity were carried out. During the calculations the following parameters were changed, namely: $\varepsilon_1 = 1 \div 10$ for the dielectric rod; the rod diameter D1 = 2mm÷5mm; the external diameter of the Teflon capillary D1 = 2mm÷5mm and the inner diameter of the capillary D2 = 0.5mm÷3mm), $\varepsilon_2 = 1 \div 80$ for the solutions. In all cases, the sample position was changed relative to the geometric center of the resonator.

It was found that the $TE_{107}$ mode is excited in the empty resonator (Fig. 2).

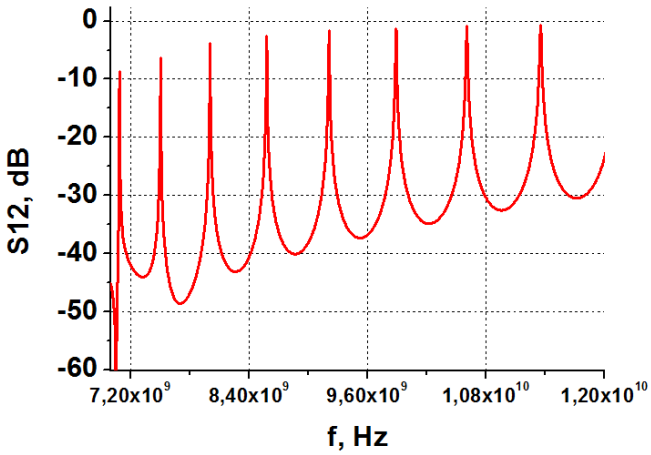


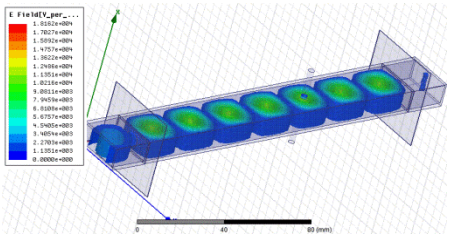

Fig. 2. The spectrum of the empty resonator

The frequency dependence of the $S_{12}$ parameter was analyzed when the dielectric permittivity of the sample was changed within the limits $1 < \varepsilon_1 < 10$. It is worth noting that a choice of the optimal sample diameter will be determined not only by the minimal perturbation of the spatial distribution of the resonant mode when the sample is placed at the antinode of the electric field component, but also by the accuracy of measuring the maximum frequency shift when the sample is moved along the resonator axis.

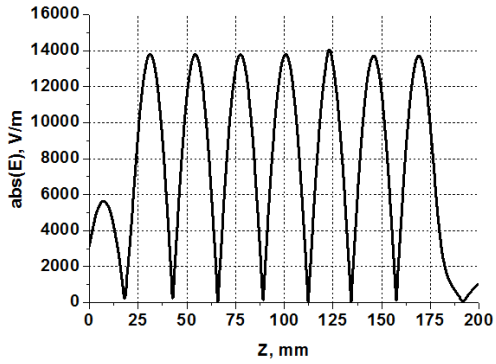
### A. Teflon sample

For the sample with a small dielectric permittivity as the Teflon, the calculated E-field distribution is shown in the Figure 3a when the sample is located in a one of the field maximum. It was found that the sample like that does not make visible changes to the field structure (Fig. 3b) up to the sample's diameter D1 = 5мм.

The dependence of the resonance frequency shift on the sample position in the standing wave field shows that for the aforementioned sample diameters, both in calculation and in experiment, the resonance frequency shift is accurately determined (Fig. 4).



a




b

Fig. 3. The calculated E-field distribution of the resonator with the Teflon rod inside (f=9.1766GHz, D1=3mm).

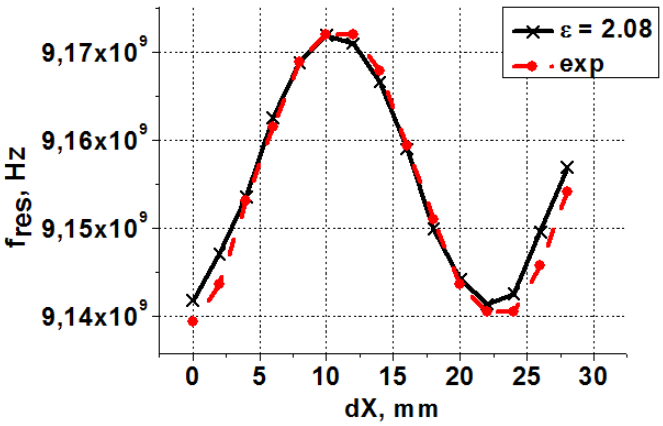



Fig. 4. Resonance frequency shift on the sample position for Teflon (ε=2.05, D1=3mm).

### B. Ruby sample

As the permittivity of the sample increases, the spatial field structure becomes more deformed. As an example, the Figure 5a shows the field pattern when the sample with ε = 10 is located at the same antinode of the field, as in the case with ε = 2.05 (see Fig. 3). It can be seen that the apparent distortion of the amplitude distribution of the E- field component at the location of this sample indicates the concentration of the SHF field inside the latter (Fig. 5b).



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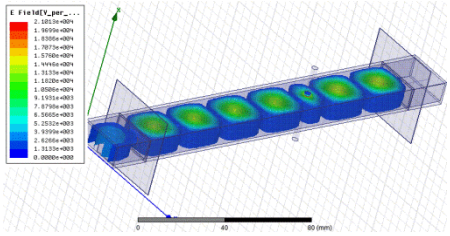

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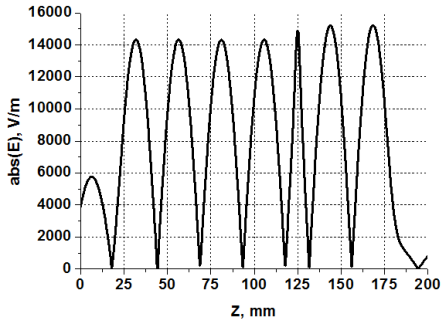



b

Fig. 5. The calculated E-field distribution of the resonator with the dielectric inside (ε=10, f=8.919 GHz)

In order to make the correct measurements, the diameter of the sample has a limitation, since, despite the confident determination of the frequencies corresponding to the position of the sample at the nodes and antinodes of the field (Fig. 6), a significant distortion of the resonance field introduced by the sample (Fig. 5) makes it more preferable to use the samples of not more than 2 mm in diameter.

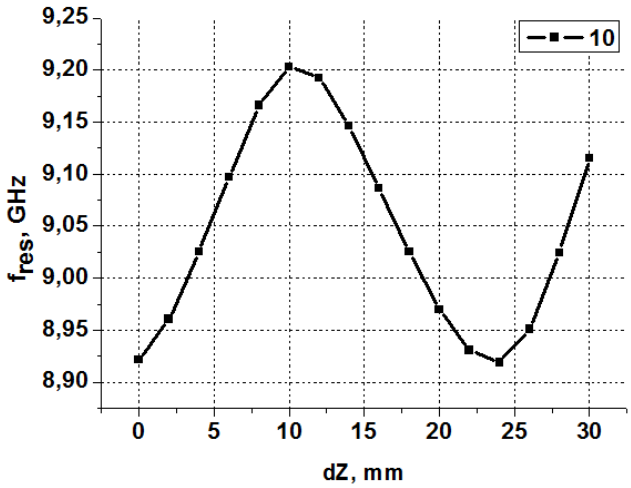
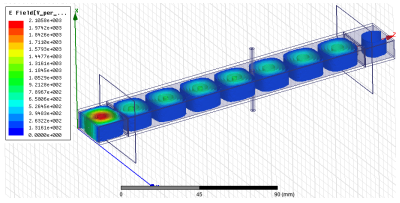



Fig. 6. Resonance frequency shift on the sample position for Ruby sample (ε=10, D1=3mm)

### C. *Teflon capillary filled with the solution*

The spatial distribution of the electric field in the resonator, when the Teflon capillary with distilled water is located both at the field minimum (Fig. 7a) and at the field maximum (Fig. 7b) show that the E-field structure is essentially varies in both cases with respect to the field distribution in an empty resonator.



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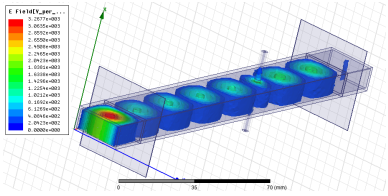

b

Fig. 7. The calculated E-field distributions in the resonator with the Ruby sample: f=9.16 GHz, z=10mm (a); f=8.86 GHz, z=24mm (b)

The Figure 8 shows the field patterns when the capillary is located at the node (z=10mm) and antinode (z=24mm) of the field. It can be seen that the apparent distortion of the amplitude distribution of the E- field component at the location of the capillary at the antinode of the field indicates the useful concentration of the SHF field inside the capillary with the solution under test.

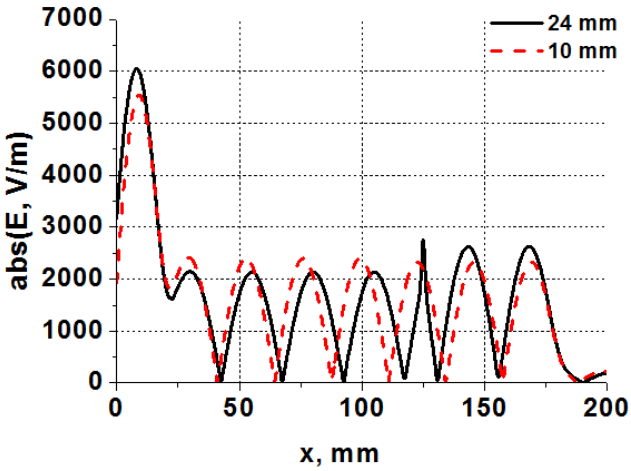


Fig. 8. The calculated E-field distributions of the resonator for two positions of the capillary

The dependence of the frequency shift of the resonator on the sample position in the standing wave field (Fig. 9) shows that the calculated and measured curves are in good agreement when using a capillary with D1 = 3mm and D2 = 1mm.



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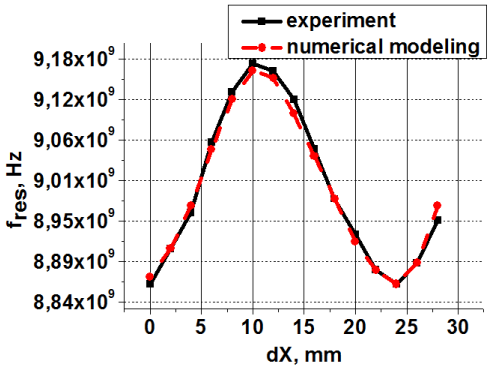


Fig. 9. . Resonance frequency shift on the position of capillary with the distilled water (D1=3mm, D2=1mm)

## IV. Conclusions

With respect to the resonance methods of dielectrometry using a rectangular X-band resonator, the results of calculations and experiments, confirming the latter, have made it possible to determine the optimum diameters of the samples (in particular, Teflon and ruby), depending on their dielectric permittivity, namely:

- for the samples with $\varepsilon < 4$, the perturbation introduced by the sample is insignificant. In these cases, the diameter of the samples can reach 5 mm;
- for the samples with $\varepsilon > 4$, the perturbation introduced by the sample is significant. Therefore, the diameter of the samples should not exceed 2 mm.

For a capillary made of Teflon with the liquid being studied, the permittivity of which is $\varepsilon > 50$, the optimal dimensions of such a capillary are as follows: the outer diameter is equal to 3 mm and the inner diameter is equal to 1 mm.

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